Much Ado about Prompting

Text Classification in Experimental Economics

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Eliciting subjects' minds, with text

Methods for eliciting reasoning in text.

Cooper & Kagel (2005)	team chat
Charness & Dufwenberg (2006)	open-ended messages
Burchardi & Penczynski (2014)	team communication protocol
Capra (2019)	think-aloud protocol
Arad, Hugh-Jones & Penczynski (2024)	chat among subjects

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Arad, Hugh-Jones & Penczynski (2024)	chat among subjects
Arad & Severin (2026)	instructions to an LLM

Classifying subjects' text

Supervised ML	Penczynski 2019; Georgalos & Hey 2020; Hüning et al. 2022
Unsupervised ML	Hanaki & Ozkes 2023; Andres et al. 2023

Supervised: needs a labelled training set; degrades as data shrinks or concepts become nuanced.

Unsupervised: emergent categories require careful, subjective interpretation.

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LLMs	this paper

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Open-ended survey data in economics Haaland, Roth, Stantcheva & Wohlfart, 2025

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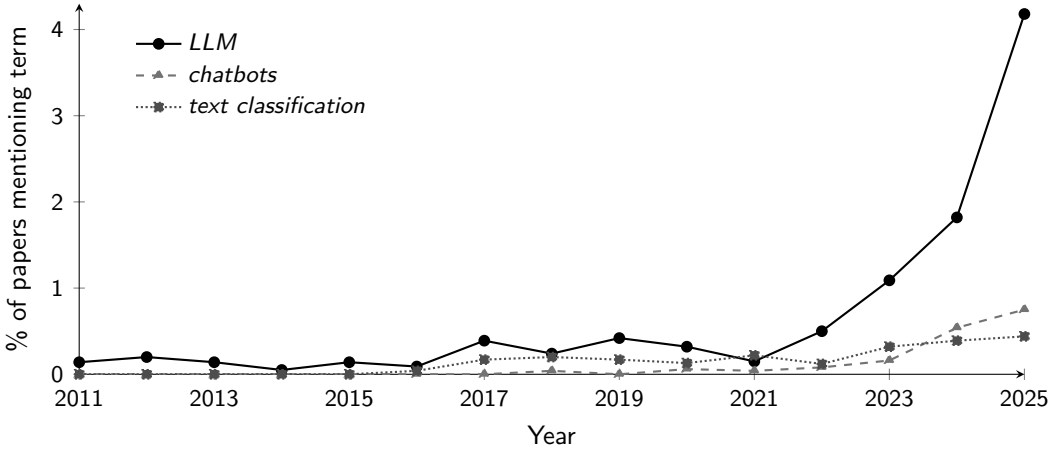
Qualitative interviews with AI	Chopra & Haaland 2024; Velez & Liu 2024; Geiecke & Jaravel 2025

Interest in top economics journals



14 journals, ∼48,000 papers, 2011–2025. Source: Gentzkow's econlit pipeline, `paulgp.com/econlit-pipeline`.

Why LLMs change the calculus

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Stronger	complex concepts — promises, level-k reasoning

Before LLMs: codebooks for humans

To classify experimental text, a researcher writes a *codebook* to train research assistants. It typically contains:

Experiment context	game mechanics, decision environment
Theory	the framework behind the categories
Category definitions	inclusion / exclusion criteria per label
Annotated examples	walk-through to cover edge cases

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Can a researcher prepare a codebook for an *LLM* the same way they would for a *human annotator*?

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And for the referee: how should a study's prompt be evaluated to judge whether the classifications can be trusted?

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Three strands of literature greet the researcher.

1. *Mostly untested advice*

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3. *Prompt brittleness*

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Follow the vendor's prompting guide	OpenAI, Anthropic, Google, Meta

Empirical results

Formatting matters	*Formatting doesn't*
Sclar 24; Han 25	Pangakis 23

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Lupo 24; Halterman & Keith 25; Atreja 25

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Reasoning hurts

Wu 23; Sprague 24

Brittle prompts

"Minor wording alterations [. . .] produce varying outputs."	Reiß 2023
"Seemingly irrelevant formatting choices produce up to 76 pp accuracy swings."	Sclar et al. 2024
"Rankings reverse across equivalent paraphrased instructions."	Mizrahi et al. 2024
"Proprietary models require explicit justification of prompts because they are so sensitive to phrasing."	Palmer & Spirling 2024

The state of prompting in social science

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Conflicting empirical results

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The researcher's options:

(a) *Avoid LLMs altogether.*

(b) *Wing it — and hope for the best.*

(c) *Pick advice from the pile — untested, perhaps misleading.*

What we promise

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- *Guide referees on what to scrutinise.*

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- *1,182 treatments* — model $\times$ variant $\times$ task.

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- *Five prompt-design dimensions*: content, format, framing, examples, reasoning.
- *1,182 treatments* — model $\times$ variant $\times$ task.
- *703,253 API calls*, 99.94 % successful, over 13 days (24 March — 6 April 2026).

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RQ3. *How sensitive is classification to how the codebook content is expressed — paraphrasing, verbosity, noise?*

RQ4. *How does model scale affect classification performance and moderate the sensitivity to prompt content and design?*

Preview of what we find

Two dimensions drive everything.

Prompt specification — is the prompt well-specified (detailed category definitions and worked examples) or underspecified (a one-line category name)?

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Brittleness reflects underspecified prompts, small models, or both.

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Each study tested a different corner of specification × capability.

Sclar 24: 76 pp format swings $\Rightarrow$ underspecified prompts, small LLaMA / Mistral

↗ what we find vs. these claims

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Rückdeschel 25: detail hurts	⇒	small open model (Llama 8B)
Mizrahi 24: paraphrase flips ranks	⇒	small models, short task descriptions

↗ what we find vs. these claims

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Same prompt, same model, same temperature — outputs still flip across runs.

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To call an effect real, we need a paired statistical test.

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Accuracy is the LLM’s agreement rate with that joint human ground truth.

McNemar's test

Same 2,063 items classified twice — under prompt A and prompt B.

	B correct	*B wrong*
A correct	a	b
A wrong	c	d

Where the two agree (a, d) tells us nothing about which prompt is better.
Where they disagree (b, c) is all the signal.

If A is truly better, more items should fall in b (A right, B wrong) than in c. McNemar tests whether $b \neq c$ beyond what random flipping predicts.

Every paired comparison in our experiments uses McNemar at $p < 0.001$.

Accuracy estimates use 95 % Wilson confidence intervals

↗ details.

Datasets

Promise	public goods, chat $\rightarrow$ promise / empty talk	719
Level-k I	voting game, team communication $\rightarrow$ level of strategic thinking 0–3	493
Level-k II	coordination game, team communication $\rightarrow$ level of strategic thinking 0–5	851

Promise: concept familiar from pre-training. *Level-k*: domain-specific theory.

A shared prompt template

```
# General Task -- Classify X in E

# Role Persona -- Act as a behavioural economist specialised in G

# Context
## Experiment -- game mechanics, payoffs, protocol
## Theory -- theoretical framework (L-tasks only)

# Classification Task -- categories, assignment rules

# Classification Coding -- output labels

# Examples -- worked examples (n-shot only)

# Constraints / # Output Format -- JSON
```

Four models

	larger	*smaller*
Proprietary	GPT 5.4	GPT 5.4-nano
Open weight	Qwen 3.5 397B (17B MoE)	Qwen 3.5 9B (dense)

Model	\$/M input	\$/M output	vs. Qwen 9B
Qwen 3.5 9B	0.10	0.15	1×
GPT 5.4-nano	0.20	1.25	∼2.5×
Qwen 3.5 397B	0.39	2.34	∼5×
GPT 5.4	2.50	15.00	∼31×

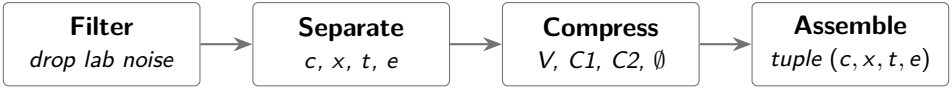
Experiment 1 · Content ablation

Decompose the codebook into three components; vary each independently.

Component	Levels
Classification instructions (c)	V, S
Experiment context (x)	V, C1, C2, $\emptyset$
Theory (t) — L-tasks only	V, C1, C2, $\emptyset$

Promise: $2 \times 4 = 8$ variants. *Level-k I*: $2 \times 4 \times 4 = 32$. *Level-k II*: 32.
72 variants $\times$ 4 models = 288 cells.

Experiment 1 · Codebook → prompt pipeline



c = classification instructions x = experimental context t = theory e = examples

All LLM-assisted steps use Claude Opus 4.6.

Experiment 1 · Compression — how much is kept

Iterative compression (V → C1 → C2) with a task-specific key-term list.

Source	Level	Cosine sim.	% words	Key terms
Experiment context	V	1.000	100	10
	C1	0.971	49	10/10
	C2	0.940	31	10/10
	∅	—	0	—
Theory	V	1.000	100	9
	C1	0.948	55	9/9
	C2	0.839	26	9/9
	∅	—	0	—
Classification instructions	V	1.000	100	—
	S	0.722	5	—

Example: Level-k II. Cosine ↗ (what is this?) from `text-embedding-3-large`. Every key term retained at every level.

Experiment 1 · Two information levels

Verbatim — the codebook, filtered but untouched. Full classification instructions (c), experimental context (x), theory (t).

(c_V, x_V, t_V, $\emptyset$) — *1,000–1,500 words.*

Minimal — a one-line description of each category. No experimental context, no theory, no examples.

(c_S, $\emptyset$, $\emptyset$, $\emptyset$) — *30–80 words depending on the task.*

Experiment 2 · Format

Same content, five visual presentations.

Format	md markers	titles	bullets	description
ml (baseline)	✓	✓	✓	`# headers` + `- bullets`
tl		✓	✓	plain titles + bullets
mp		✓		plain titles + prose
fl			✓	no headers, bullets
pt				no headers, prose

4 alternative formats × 2 information levels (minimal, verbatim) = *8 new variants / task* × 3 tasks × 4 models = *96 cells*.

Plus two system-vs-user message splits on verbatim content.

↗ scatters · text example

Experiment 3 · Framing

Two framing elements, tested by removal.

Element	Example
General task (GT)	"Classify messages from a coordination game."
Role persona (RP)	"You are a behavioural economist specialised in level-k."

Three removal conditions: *no RP*, *no GT*, *neither*. Compared against the full-framing baseline.
Each at both information levels (minimal, verbatim). 3×3 tasks $\times$ 2 levels = *18 comparisons / model.*

Experiment 4 · Examples, and how to present them

Examples. n-shot from the codebook's worked examples.

P: 21 ($\sim$10/label) L_1: 13 ($\sim$3/label) L_2: 5 ($\sim$1/label)

Tested at *minimal* and *verbatim* content — separates content vs. examples as sources of information.

Presentation styles. Five:

markdown (baseline) · *XML tags* · *delimiter* · *arrow* · *multi-turn*

First four in the system message; *multi-turn* = user/assistant dialogue outside the system message.

↗ see the five styles

Experiment 5 · Reasoning

Enable model reasoning at multiple effort levels.

Models	Levels tested	Median tokens
GPT 5.4 / nano	low, medium, high, extra-high	113 / 309 / 682 / 2,479
Qwen 397B / 9B	on / off	∼750 / 0

Crossed with content (verbatim / minimal) × shot (0-shot / *n*-shot) — tests whether reasoning can compensate for reduced prompt detail, and whether examples absorb reasoning gains.

Experiment 6 · Information fidelity (1/2)

Hold the informational content approximately constant, vary the surface expression.

Treatment	Length change	Cosine to verbatim
Verbose	+50 to 275 % words, relevant elaboration	0.87–0.97
Noise	matched length, irrelevant padding (red-squirrel facts)	0.57–0.73

Each treatment is applied at 3 levels — *classification instructions only*, *experimental context only*, or *both*:

Verbose: 3 levels × 3 tasks = 9 comparisons / model. *Noise*: 3 levels × 3 tasks = 9 comparisons / model.

Total: *18 comparisons / model* at each shot condition (0-shot and *n*-shot).

The contrast between *verbose* and *noise* isolates *informational relevance* from *prompt length*.

↗ verbose & noise example

Experiment 6 · Information fidelity (2/2)

Two further tests of expression-vs-content:

Treatment	What changes
Paraphrase	round-trip back-translation: EN $\rightarrow$ TR $\rightarrow$ EN and EN $\rightarrow$ DE $\rightarrow$ EN meaning preserved (cosine 0.97–0.99); surface wording varies
Raw codebook	original document with payment instructions, software notes, formatting artefacts — no filter, no compression

Paraphrase: 2 paraphrases $\times$ 3 tasks $=$ 6 comparisons / model.
Raw codebook: one head-to-head against the cleaned pipeline output (n-shot).

Results

↗ a note on scope

Content · the two information levels

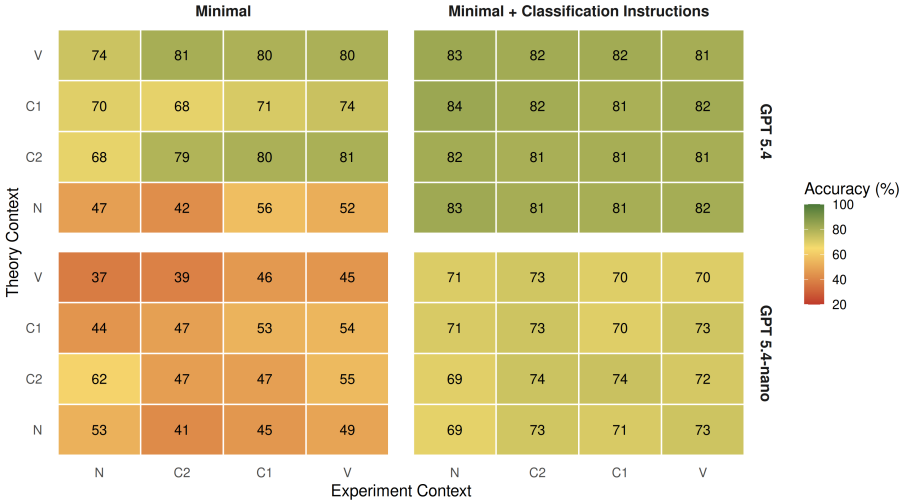
	Minimal			*Verbatim*		
	P	L_1	L_2	P	L_1	L_2
GPT 5.4	**81.0**	47.4	49.6	**81.6**	81.3	**81.0**
GPT 5.4-nano	72.4	**52.5**	**53.1**	75.0	70.2	68.9
Qwen 397B	79.1	50.1	43.6	76.1	**83.0**	74.6
Qwen 9B	74.6	**53.1**	38.8	74.1	70.8	64.4

Bold = best in each column, and any value statistically tied with it.

Promise: done already at minimal. Level-k: jumps once content is there.

↗ what drives this

Content · information surface (L_1)



↗ 4-model surfaces

Result 1

- *Context and theory partially substitute for missing instructions.*
- *Larger models benefit from more detailed content.*
- *Smaller models perform better with less.*

Format & framing · proportion significant

	Format				*Framing*			
	Minimal		Verbatim		Minimal		Verbatim	
	+	−	+	−	+	−	+	−
GPT 5.4	**.17**	.08	.00	.08	**.33**	**.11**	**.33**	.00
GPT 5.4-nano	.00	.00	.08	.00	**.33**	**.22**	.00	.00
Qwen 397B	.08	.08	.00	.00	**.11**	**.33**	.00	.00
Qwen 9B	**.25**	**.17**	.00	.00	**.78**	.00	.00	**.11**

On well-specified prompts: almost nothing is significant. On underspecified prompts: inconsistent direction.

↗ scatter plots · format example

Result 2

- *Neither formatting nor framing reliably affects classification accuracy.*
- *On verbatim prompts, effects are negligible.*
- *On minimal prompts, significance increases but the direction is inconsistent across models and tasks.*

Examples · content and examples substitute

GPT 5.4; the pattern is similar for the other three models.

	P	L_1	L_2
Δ_{Ex}	$+11.7^*$	$+37.4^*$	$+20.4^*$
$\Delta_{\text{Ex}} - \Delta_{\text{Content}}$	$+11.1^*$	$+3.4$	-10.9^*
$\Delta_{\text{Ex} \mid \text{Content}}$	$+9.9^*$	$+2.6$	$+0.7$
$\Delta_{\text{Content} \mid \text{Ex}}$	-1.2	-0.8	$+11.6^*$

Δ_{Ex}: examples added to minimal. $\Delta_{\text{Ex}} - \Delta_{\text{Content}}$: examples alone vs. content alone.
$\Delta_{\text{Ex} \mid \text{Content}}$: examples given verbatim content. $\Delta_{\text{Content} \mid \text{Ex}}$: content given *n*-shot.

Examples dominate on familiar concepts; content dominates on theory-dependent tasks.

Result 3

- *Content and examples carry overlapping information — partial substitutes.*
- *Examples dominate on recognition-heavy tasks.*
- *Content dominates on theory-dependent tasks.*

Prompt $\approx$ codebook

	P	L_1	L_2
GPT 5.4	**91.2**	**84.0**	**81.7**
GPT 5.4-nano	79.4	78.3	70.7
Qwen 397B	**90.0**	**83.6**	77.8
Qwen 9B	84.1	**82.8**	68.2

Verbatim + n-shot, no reasoning. Bold = best in the column, and any value statistically tied with it.

Example format

	In-prompt formats				*Multi-turn*			
	Minimal		Verbatim		Minimal		Verbatim	
	+	−	+	−	+	−	+	−
GPT 5.4	.00	.00	.00	.00	.00	**.33**	.00	**.33**
GPT 5.4-nano	.00	.00	.00	.00	.00	**.67**	.00	.00
Qwen 397B	.00	**.33**	.00	**.33**	.00	**.67**	.00	**.33**
Qwen 9B	**.22**	**.33**	.00	**.33**	.00	**.67**	.00	**.33**

In-prompt format: indifferent. Examples outside the system message: consistently hurts.

↗ the five example styles

Result 4

- *How examples are formatted within the system prompt does not matter.*
- *Placing examples outside the system prompt (as multi-turn conversation) consistently hurts all models.*

Reasoning · can it substitute for content?

GPT 5.4; the pattern is similar for the other three models.

	P	L_1	L_2
Δ_{Rea}	$+6.6^*$	-11.5^*	$+19.7^*$
$\Delta_{\text{Rea}} - \Delta_{\text{Content}}$	$+5.9^*$	-45.4^*	-11.6^*
$\Delta_{\text{Content} \mid \text{Rea}}$	-6.8^*	$+46.7^*$	$+11.6^*$
$\Delta_{\text{Rea} \mid \text{Content}}$	-0.8	$+1.2$	$+0.0$

Δ_{Rea}: reasoning added to minimal. $\Delta_{\text{Rea}} - \Delta_{\text{Content}}$: reasoning alone vs. content alone.
$\Delta_{\text{Content} \mid \text{Rea}}$: content given reasoning is on. $\Delta_{\text{Rea} \mid \text{Content}}$: reasoning given content is on.

Content keeps its full value when reasoning is on; reasoning adds nothing once content is present.

Reasoning · effect at 0-shot vs. *n*-shot

	0-shot			*n-shot*		
	P	L_1	L_2	P	L_1	L_2
GPT 5.4	−0.8	+1.2	0.0	+0.8	+2.0	+1.9
GPT 5.4-nano	+4.9*	+5.5	+1.6	+8.3*	+1.0	−0.4
Qwen 397B	+5.9*	+1.0	+0.6	+2.5*	+2.8	+2.6
Qwen 9B	+8.6*	+10.5*	+2.4	+6.5*	+1.6	+4.0

*pp change vs. no reasoning. GPT: high effort; Qwen: on. * = significant.*

Frontier: no gain. Smaller models: P-gains; L-gains absorbed by examples.

Result 5

- *Reasoning does not improve classification for frontier models on any task.*
- *Smaller models benefit on recognition-heavy tasks; gains disappear once examples are present.*
- *Reasoning cannot substitute for codebook content. Content alone outperforms reasoning alone on L-tasks.*
- *Reasoning and content are complements, not substitutes — reasoning amplifies what the model already knows.*

Cost of reasoning

Reasoning effort	Cost ratio	Latency
Low	0.38× prompt	2.5×
Medium	1.09× prompt	4.7×
High	2.53× prompt	8.1×
Extra-high	8.42× prompt	21×

GPT 5.4 Azure pricing, April 2026.

Information fidelity · verbose and noise

	0-shot				*n-shot*			
	Verbose		Noise		Verbose		Noise	
	+	−	+	−	+	−	+	−
GPT 5.4	.00	.00	.00	**.11**	.00	.00	.00	.00
GPT 5.4-nano	**.11**	.00	.00	**.11**	.00	.00	.00	**.11**
Qwen 397B	**.44**	.00	**.22**	.00	.00	**.22**	.00	**.33**
Qwen 9B	**.44**	.00	.00	**.11**	.00	**.11**	.00	**.33**

Proportion of statistically significant comparisons vs. the verbatim baseline. 9 per cell (3 scopes × 3 tasks).

GPT insensitive. Qwen responsive at 0-shot; at n-shot the verbose gains disappear — examples absorb the extra information.

↗ mean $\bar{\Delta}$ & σ · figures · example text

Information fidelity · paraphrase and raw codebook

	Paraphrase (sig/6)		*Raw codebook*
	0-shot	*n*-shot	(mean $\bar{\Delta}$, sig/3)
GPT 5.4	1/6	0/6	−0.5 pp, 0/3
GPT 5.4-nano	0/6	**2/6**	**−3.0 pp, 2/3**
Qwen 397B	1/6	**2/6**	**−2.8 pp, 2/3**
Qwen 9B	**2/6**	**2/6**	**−6.8 pp, 2/3**

Paraphrase: 6 comparisons / model per shot condition (2 paraphrases × 3 tasks). Median $|\bar{\Delta}| < 1$ pp throughout.
Raw codebook: *n*-shot, one head-to-head per task vs. the cleaned pipeline output.

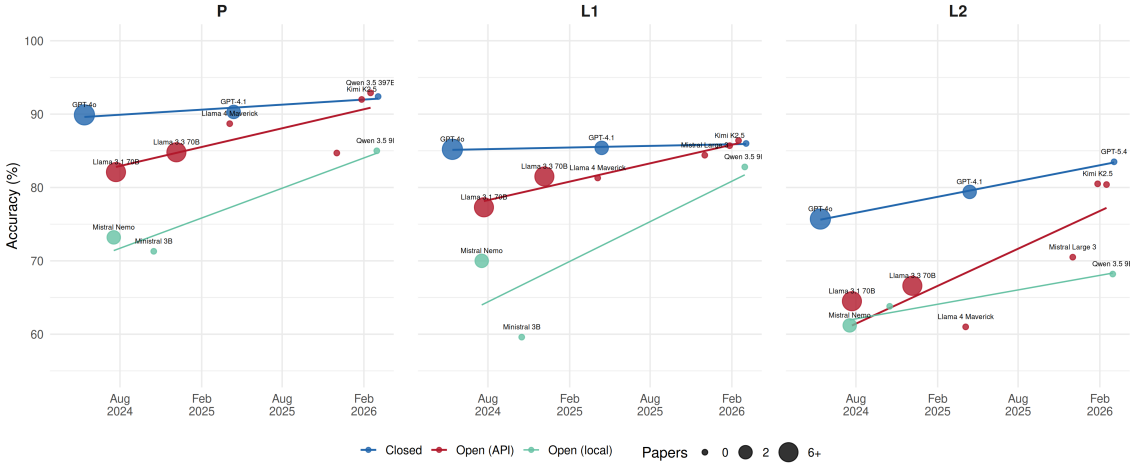
Frontier robust to both. Smaller / open models: paraphrase rarely flips, raw codebook hurts.

↗ per-task effects

Result 6

- *Frontier models are insensitive to paraphrasing, verbosity, and noise.*
- *Raw codebook on the frontier: no significant difference from the cleaned version.*
- *Raw codebook on smaller / open models: significantly negative — the extra irrelevant content hurts.*

Open or closed?



Open has closed the accuracy gap, not the sensitivity gap.

Recommendations · for the researcher

Provide a detailed codebook, the same kind you would give a human annotator.

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Role persona is fine if you want it. No reliable gain on frontier models, unpredictable on small / open models.

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Read what the prompt contains, not how it's packaged.

Ask: are categories clearly defined?

Ask: are worked examples included?

Don't reward markdown or personas.

Don't demand paraphrase robustness checks on well-specified prompts.

Flag underspecified prompts on small / open models — expect instability there.

Thank you.

can-celebi.github.io/prompting

Can Çelebi · can.celebi@univie.ac.at
Stefan P. Penczynski · s.penczynski@uea.ac.uk

Much Ado about Prompting

Content · what drives accuracy

	P	L_1	L_2
Classification instructions	2.6 %	**58.3 %**	27.4 %
Theory context	—	8.5 %	9.6 %
Experiment context	3.1 %	1.3 %	1.9 %
Model choice	**94.3 %**	31.9 %	**61.1 %**

On L: classification definitions dominate. On P: model choice does.

↗ per-model decomposition

↩ back

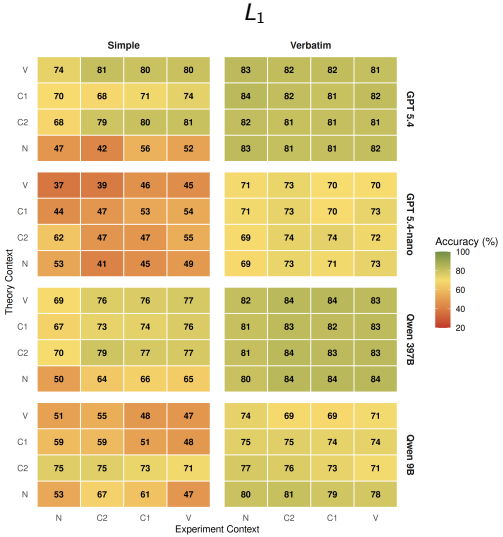
Per-model variance decomposition

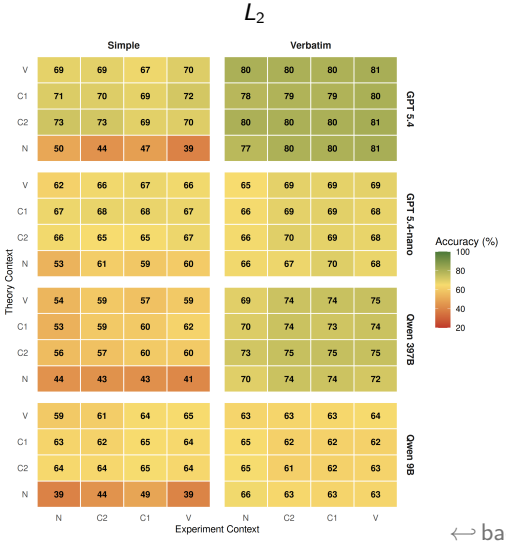
	P		L_1			L_2		
	Cls	Exp	Cls	Theory	Exp	Cls	Theory	Exp
GPT 5.4	20.6	79.4	57.4	41.6	0.9	69.7	30.2	0.1
GPT 5.4-nano	91.0	9.0	99.9	0.1	0.0	39.2	40.7	20.1
Qwen 397B	9.0	91.0	70.3	16.3	13.4	85.5	12.2	2.3
Qwen 9B	99.9	0.1	97.9	1.2	0.9	26.1	73.8	0.0

Unique explained variance (%), Type-III LR tests.

↩ back

L_1 and L_2 surfaces · all four models





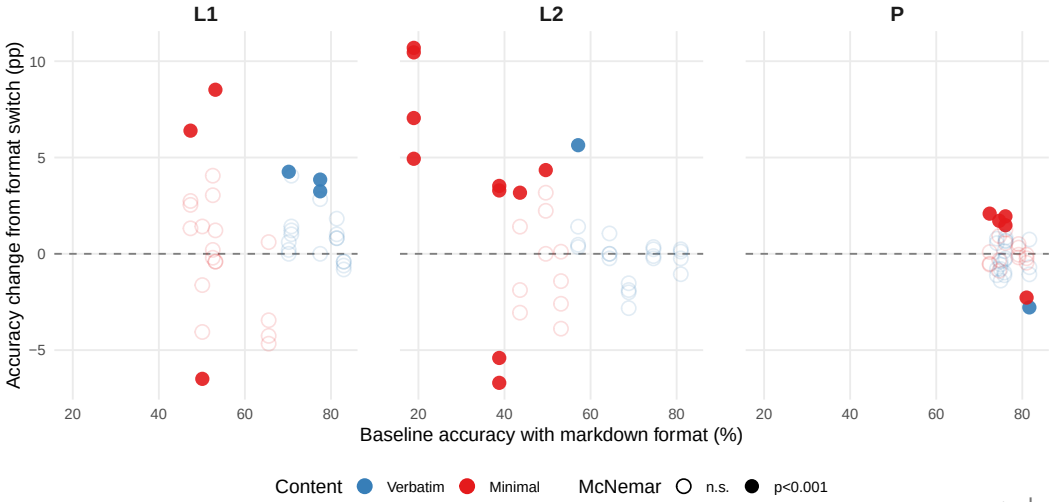
↩ back

The five format variants

```
ml (markdown + list)                                                  baseline
## Level 0
- chooses randomly or without strategic reasoning

tl (titles + list)                                                     Level 0
- chooses randomly or without strategic reasoning

mp (titles + prose)                                                    Level 0
Chooses randomly or without strategic reasoning.

fl (flat list)                                                [header removed]
- chooses randomly or without strategic reasoning

pt (plain text)                                               [header removed]
Chooses randomly or without strategic reasoning.
```

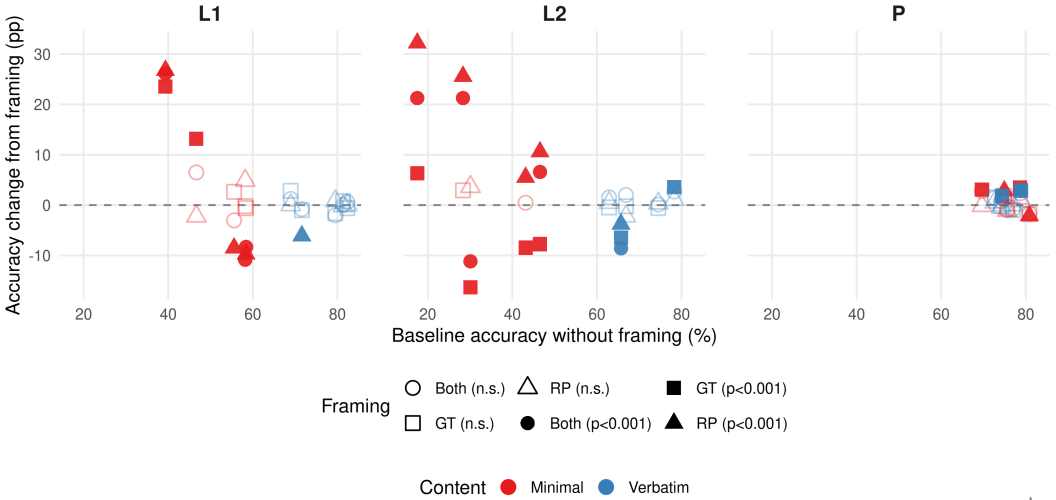
↩ back

Format effects by task



↩ back

Framing effects by task



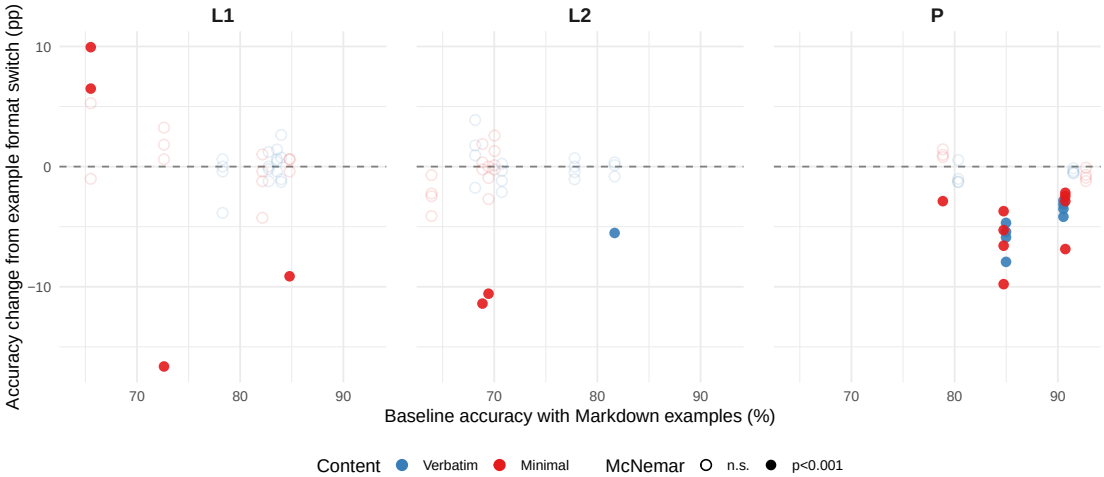
↩ back

The five example-presentation styles

```
n (markdown)                                                    ## Example 1
### Message -- "P1:  all 200 then?  P2:  yes"
### Classification -- {"p1":"0", ...}

nx (XML)                <example> <message>...</message> ...</example>

nd (delimiter)            Example 1:  Message:  ... Classification:  {...}

na (arrow)                "P1:  all 200 then?  P2:  yes" → {"p1":"0",...}

t (multi-turn)  sent as user / assistant turns, outside the system message
```

↩ back

Example format dispersion



↩ back

Reasoning — step-by-step levels (GPT models)

		0-shot			*n*-shot		
	Step	P	L_1	L_2	P	L_1	L_2
GPT 5.4	off→low	+4.2*	+3.9*	+4.4*	+0.8*	+0.4	−0.8
	low→med	0.0	+1.0	+2.4	+0.4	+2.0	+1.9
	med→high	−4.1*	−1.4	−0.5	0.0	−0.6	+2.0
	high→xhigh	−3.9*	0.0	+0.8	−0.2	−1.4	−0.6
GPT nano	off→low	+4.5*	+11.5*	+1.3*	+4.5*	+1.3	−3.4*
	low→med	+0.9	+5.1	+0.2	+3.0*	+0.6	+3.4
	med→high	0.0	−1.2	+2.5	+0.8	−0.2	+0.4
	high→xhigh	−1.4	+4.7	+6.1*	+2.0*	+3.7	+8.3*

* = *McNemar* $p < 0.001$. *Each step relative to the previous level.*

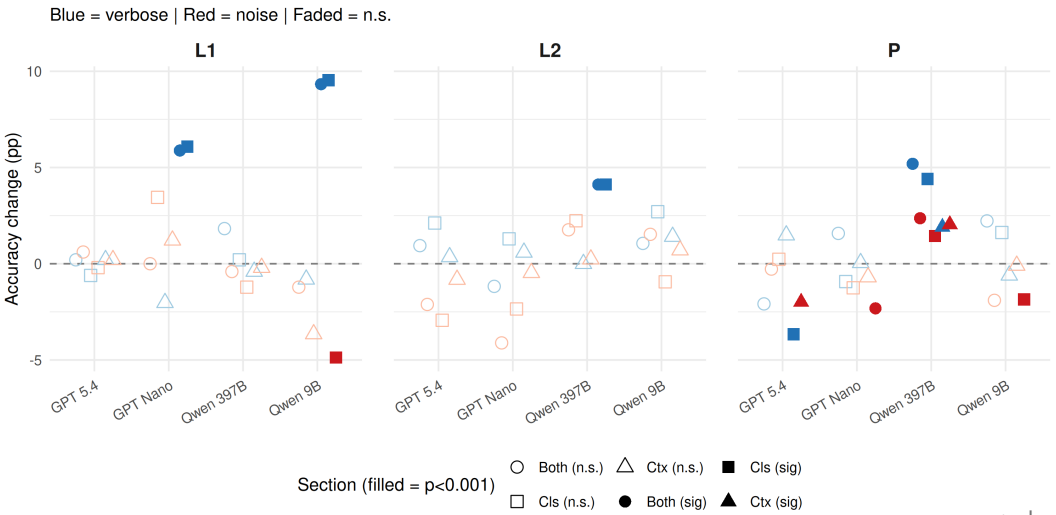
↩ back

Verbose vs. noise — an example

```
Normal (verbatim)
- Participants can invest between 0-200 in a group project.
- The total invested is doubled and shared equally.

Verbose (relevant elaboration, +50-275% words)
- Participants can invest between 0-200 in a group project, any integer.
- The total invested is doubled and shared equally.  Investing 100 returns
only 66 to the individual -- a net loss of 34, but the group gains 200.

Noise (irrelevant padding, matched length)
- Participants can invest between 0-200 in a group project.
- An adult red squirrel weighs 250-340 grams, body length 19-23 cm plus
tail 15-20 cm.
- The total invested is doubled and shared equally.
- Red squirrels build nests called dreys, 6-12 m above ground, using
twigs, moss, stripped bark.
```

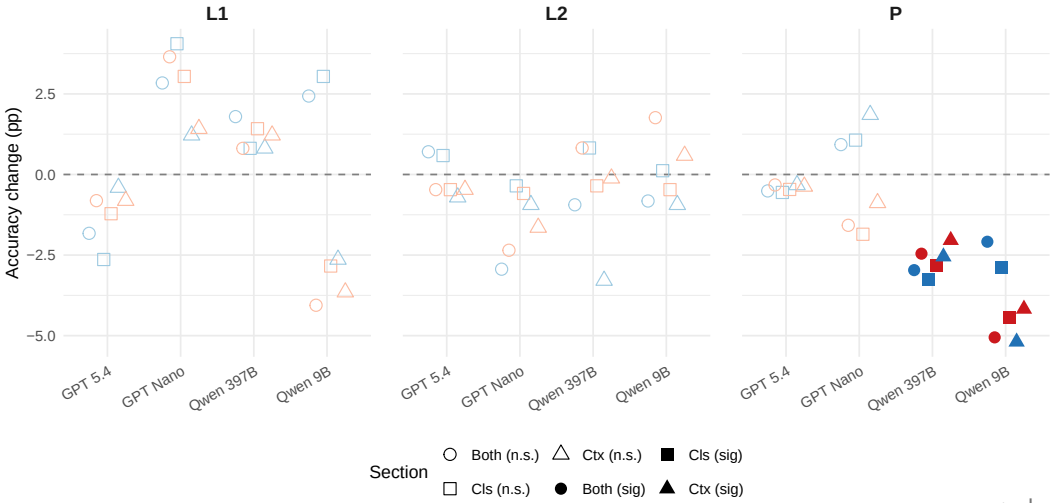
↩ back

Verbose & noise (0-shot)



↩ back

Verbose & noise (n-shot)



↩ back

Verbose & noise · mean $\bar{\Delta}$ and σ

	0-shot				*n-shot*			
	Verbose		Noise		Verbose		Noise	
	$\bar{\Delta}$	σ	$\bar{\Delta}$	σ	$\bar{\Delta}$	σ	$\bar{\Delta}$	σ
GPT 5.4	-0.1	1.8	-0.8	1.2	-0.6	1.1	-0.6	0.3
GPT 5.4-nano	$+1.4$	2.9	-0.8	2.2	$+0.9$	2.1	-0.2	2.3
Qwen 397B	$+2.3$	2.1	$+0.9$	1.3	-0.9	2.0	-0.4	1.6
Qwen 9B	$+3.1$	3.7	-1.1	2.0	-0.9	2.6	-2.5	2.5

In pp vs. verbatim baseline. 9 comparisons per cell.

↩ back

A note on scope

Experimental-economics text is a best-case setting for codebook-based classification.

- *Tightly constrained.* Fixed rules, payoffs, terms; edge cases rare. Unlike tweets, parliamentary speeches, Reddit threads.
- *Full context available.* Both *experiment context* and *formal theory* can be provided. Most social-science codebooks have neither.
- *Codebook shape differs elsewhere.* Political science / sociology: often many labels, short definitions, multi-label overlap.
- *First study* to vary experiment and theory context independently, and to cross them with formatting, framing, examples, reasoning.

Most directly relevant to *experimental economists*, then *economists using theory-based labels* for text classification.

Beyond that, generalisation needs care — a limitation of this study.

↩ back

Cosine similarity

The angle between two embedding vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, ignoring their magnitudes:

$$\cos(\mathbf{a}, \mathbf{b}) \;=\; \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \;=\; \frac{\sum_i a_i b_i}{\sqrt{\sum_i a_i^2}\,\sqrt{\sum_i b_i^2}}$$

- Range: $[-1, 1]$. *1* = same direction, *0* = orthogonal, *−1* = opposite.
- Measures *directional agreement*, not semantic equivalence — "I love it" and "I hate it" can still score high.
- Works well for *topical* / *structural* similarity; weaker on fine sentiment or stance.
- In our pipeline: cos ≥ 0.80 between verbatim and any compressed variant (`text-embedding-3-large`).

↩ back

Wilson confidence intervals

For a proportion $\hat{p} = x/n$ with $z = z_{1-\alpha/2}$:

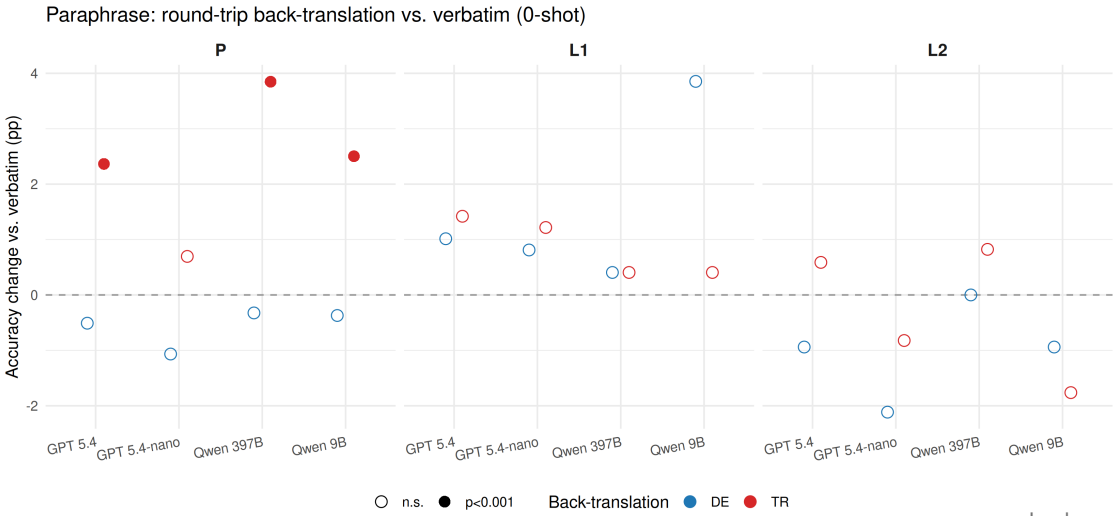
$$\hat{p}_{\mathrm{W}} = \frac{\hat{p}+\frac{z^2}{2n}}{1+\frac{z^2}{n}} \pm \frac{z}{1+\frac{z^2}{n}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}$$

- Score-based; does not require $\hat{p}$ far from 0 or 1.
- Better coverage than the Wald (normal-approximation) interval when n is small or $\hat{p}$ is near the boundary.
- Bounded inside $[0, 1]$ by construction.

Wilson (1927), *JASA*.

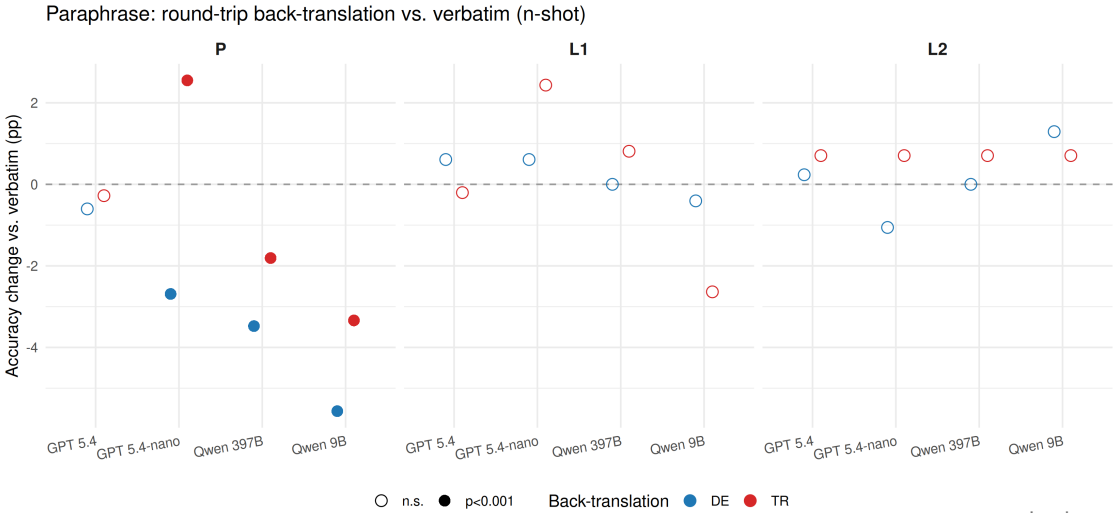
↩ back

Paraphrase effects by model and task (0-shot)



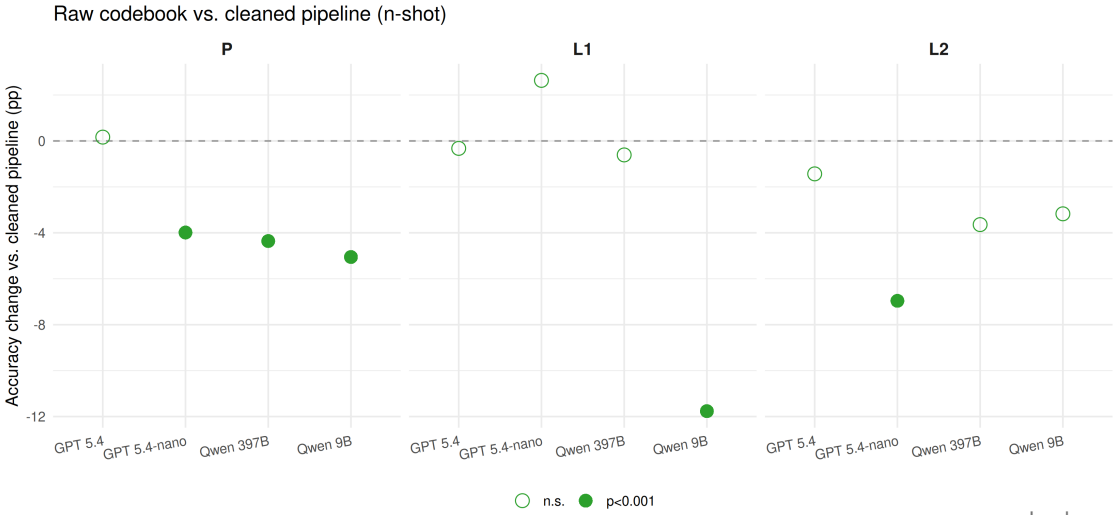
↩ back

Paraphrase effects by model and task (*n*-shot)



↩ back

Raw codebook effects by model and task



↩ back

Where we land

The prompt-engineering literature vs. what we actually find.

	Prior claim	*Our result*
Formatting	Sclar 24: +76 pp swings	well-specified: <1 pp; underspecified: scattered, no direction
Paraphrase	Reiß 23: flips on paraphrase	frontier: <1 pp; smaller models: mixed direction
Detail	Lupo 24: +21 pp; Rückdeschel 25: hurts small	helps large; too much context hurts small — both right
Persona	Törnberg 24; Dunivin 25: helps	well-specified: null; underspecified: mixed, no reliable gain
Reasoning	Dunivin 25: helps	frontier: no gain; small: helps recognition only, absorbed by examples

Prior claims hold in a specific corner — underspecified prompts on small / open models. On well-specified prompts and capable models the effects disappear.

↩ back